

MATH9102 Fundamentals of Data Analysis

Continuous Assessment

Due Date: Friday November 7th, 2025 @ 23:59 (via Brightspace)

Contributes 40% to overall module marks

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The Task

- Choose a dataset that will allow you to complete the assignment.
 - You may use a dataset listed in the document MATH9102 Locating Dataset for CA.pdf or find your own.
 - Make sure it contains a mix of interval/ratio and categorical variables to fulfil the assignment requirements or allows for these to be generated.
- Identify FOUR testable hypotheses.
 - **1 hypothesis** where both variables are interval/ratio.
 - **1 hypothesis**: interval/ratio variable compared across a **binary categorical** variable.
 - **1 hypothesis**: interval/ratio variable compared across a **multi-category nominal** variable.
 - **1 hypothesis**: two nominal variables.
- Inspect and report on your dataset suitability for data analysis considering only the concepts of interest.
- For each hypothesis:
 - State the hypothesis pair clearly.
 - Inspect and describe the variables.
 - Decide the correct statistical test (justify your choice presenting clear evidence to support it).
 - Run the test in R.
 - Report the results.
 - Draw a short conclusion.
- Code requirements
 - All work must be reproducible in **Quarto (.qmd)**.
 - Code should be clearly commented and labelled.
 - Show all data preparation, exploration and analysis steps.

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What do I need to Submit?¹

Submit a single .zip (or .rar) file containing:

- **Your dataset:**
 - Either the dataset file itself OR a link (include the link in your report).
- **Your report**
 - Can be Word (.docx) or Quarto/HTML (.html).
 - Filename must start with your student number (e.g., D1234567.html).
- **Your Quarto file(s):**
 - Containing all R code needed to reproduce your work including code you used to explore the concepts of interest to support your decision making.
 - Must start with your student number (e.g., D1234567.qmd).

How do I submit?²

- You must submit using the assignment named **CA Phase ONE** in Brightspace.
 - Navigate to Assessment → Assignments

How will I be marked?³

A **rubric** is provided in Brightspace (also published with this assignment specification).

Your grade will be based on:

- Quality and appropriateness of hypotheses.
- Correct choice of statistical tests based on valid evidence.
- Correct execution of tests.
- Clear communication of results and conclusions (tables, graphs, writing).
- Quality, readability, and reproducibility of your code.

Feedback will be given using the rubric categories.

Important Notes

- You must adhere to the conventions used in examples provided in this module. Deviations will result in deduction of marks.
- Your report **must** be self-contained. If it is not, marks will be deducted.
- You **cannot resubmit** your work once it has been marked – no second submissions will be accepted.
- **Advice and guidance** on the assignment will be provided in lectures, labs.
- Submissions sent by email before the deadline asking for review, correction, or pre-marking **will not be** read or accepted.
 - You can ask questions by email. Questions should be specific and not be asking for pre-marking or correction in advance.
- Make sure your work is complete, correct, and ready before you upload it to Brightspace.

¹ You **cannot** submit a Python script. If you do it will be treated as a non-submission, and you will receive a mark of 0.

² Only submissions made using Brightspace will be marked. Email submissions will be ignored.

³ You will not receive actual marks until the module results have been reviewed by the external examiners and processed at the examination board in January 2026. You will receive clear indicators of your level of achievement.